

Chapter 2: Data, measurements, and data preprocessing

"Without data, you're just another person with an opinion." – W. Edwards Deming

Data Mining: Concepts and Techniques

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Chapter 1

What is data mining?

KDD process steps

Applications

Image: Flowchart showing KDD process with Step 2 (Data Preprocessing) highlighted

Chapter 2

What is data?

Deep dive into data preparation

Data quality & preprocessing

From Data Mining Process to Data Itself

• [Linking to Previous Chapter](#)

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Chapter Roadmap

Data Types	Nominal, binary, ordinal, numeric
Statistics of Data	Mean, variance, correlation, visualizations
Similarity Measures	Distance metrics, cosine, KL divergence
Data Quality & Cleaning Data	Missing values, noise, integration
Transformation	Normalization, discretization, compression
Dimensionality Reduction	PCA, attribute selection, nonlinear methods

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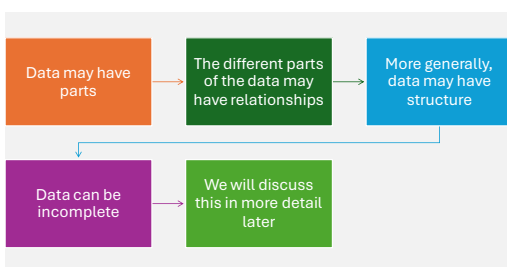
What is Data?

- Collection of **data objects** and their **attributes**
- An **attribute** is a property or characteristic of an object
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, dimension, or feature
- A collection of attributes describe an **object**
 - Object is also known as record, point, case, sample, entity, or instance

Attributes					
Tid	Refund	Marital Status	Taxable Income	Cheat	
1	Yes	Single	125K	No	Objects
2	No	Married	100K	No	
3	No	Single	70K	No	
4	Yes	Married	120K	No	
5	No	Divorced	95K	Yes	
6	No	Married	60K	No	
7	Yes	Divorced	220K	No	
8	No	Single	85K	Yes	
9	No	Married	75K	No	
10	No	Single	90K	Yes	

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A More Complete View of Data



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Attribute Values

Attribute values are numbers or symbols assigned to an attribute for a particular object

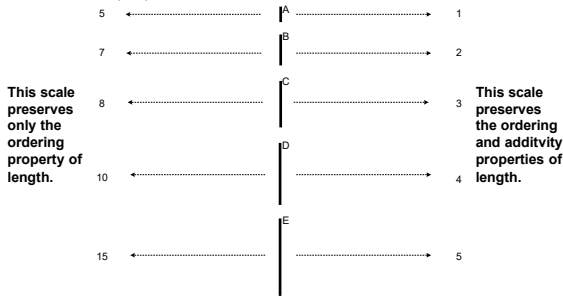
Distinction between attributes and attribute values

- Same attribute can be mapped to different attribute values
 - Example: height can be measured in feet or meters
- Different attributes can be mapped to the same set of values
 - Example: Attribute values for ID and age are integers
- But properties of attribute values can be different

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Measurement of Length

- The way you measure an attribute may not match the attributes properties.



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Types of Attributes

- There are different types of attributes
 - Nominal
 - Examples: ID numbers, eye color, zip codes
 - Ordinal
 - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height {tall, medium, short}
 - Interval
 - Examples: calendar dates, temperatures in Celsius or Fahrenheit.
 - Ratio
 - Examples: temperature in Kelvin, length, time, counts

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Nominal Attributes

Properties:

- No order (Brown $\neq$ Black)
- Can be coded as numbers but no math!
- Central tendency = **Mode**

Example:

Survey data: Hair color codes {1=Black, 2=Brown, 3=Blond}. Code 2 $\neq$ 2 $\times$ code 1.

Attribute	Possible Values
Hair color	{Black, Brown, Blond, Red}
Marital status	{Single, Married, Divorced}
Occupation	{Teacher, Engineer, Doctor}

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Binary Attributes : Only Two States: 0 or 1

Binary is like a light switch—unless it's a broken switch (asymmetric).

Type	Meaning	Example
Symmetric	Both states equally important	Gender {M, F}
Asymmetric	One state more important (rare)	HIV Test {Positive=1, Negative=0}

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Ordinal Attributes Ordered but Not Measured

Properties:

- Order matters (Large $>$ Medium)
- But difference unknown (Large - Medium $\neq$ Medium - Small)
- Central tendency = **Median, Mode**

Example:

Survey: Satisfaction 1–5. Can say 5 $>$ 3, but not “5 is twice as good as 2.5.”

Attribute	Ordered Values
Drink size	{Small, Medium, Large}
Satisfaction	{Very Dissatisfied, ..., Very Satisfied}
Education	{High School, Bachelor, Master, PhD}

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Numeric Attributes

Example:

- Interval: 20°C - 10°C = 10°C OK, but 20°C is not twice as hot as 10°C.
- Ratio: \$100 is twice \$50.

Measurable Quantities

Type	Zero Point	Example	Math Operations
Interval-scaled	Arbitrary (no true zero)	Temperature (°C), Dates	Differences OK, ratios not
Ratio-scaled	True zero	Height, Weight, Income	Differences & ratios OK

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Properties of Attribute Values

- The type of an attribute depends on which of the following properties/operations it possesses:

- Distinctness: $= \neq$
- Order: $< >$
- Differences are meaningful: $+ -$
- Ratios are meaningful: $* /$

- Nominal attribute: distinctness
- Ordinal attribute: distinctness & order
- Interval attribute: distinctness, order & meaningful differences
- Ratio attribute: all 4 properties/operations

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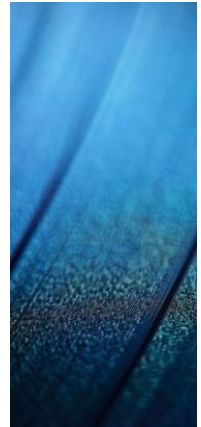
Difference Between Ratio and Interval

Is it physically meaningful to say that a temperature of 10 ° is twice that of 5° on

- the Celsius scale?
- the Fahrenheit scale?
- the Kelvin scale?

Consider measuring the height above average

- If Bill's height is three inches above average and Bob's height is six inches above average, then would we say that Bob is twice as tall as Bill?
- Is this situation analogous to that of temperature?



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Attribute Type	Description	Examples	Operations
Nominal	Nominal attribute values only distinguish. ($=, \neq$)	zip codes, employee ID numbers, eye color, sex: { <i>male, female</i> }	mode, entropy, contingency correlation, χ^2 test
Ordinal	Ordinal attribute values also order objects. ($\leq, >$)	hardness of minerals, { <i>good, better, best</i> }, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Interval	For interval attributes, differences between values are meaningful. ($+, -$)	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, t and F tests
Ratio	For ratio variables, both differences and ratios are meaningful. ($*, /$)	temperature in Kelvin, monetary quantities, counts, age, mass, length, current	geometric mean, harmonic mean, percent variation

- This categorization of attributes is due to S. S. Stevens

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Attribute Type	Transformation	Comments
Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
Ordinal	An order preserving change of values, i.e., $new_value = f(old_value)$ where f is a monotonic function	An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by { 0.5, 1, 10}.
Interval	$new_value = a * old_value + b$ where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
Ratio	$new_value = a * old_value$	Length can be measured in meters or feet.

- This categorization of attributes is due to S. S. Stevens

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2.1.5 Discrete vs. Continuous Attributes

Discrete Attribute

- Has only a finite or countably infinite set of values
- Examples: zip codes, counts, or the set of words in a collection of documents
- Often represented as integer variables.
- Note: binary attributes are a special case of discrete attributes

Continuous Attribute

- Has real numbers as attribute values
- Examples: temperature, height, or weight.
- Practically, real values can only be measured and represented using a finite number of digits.
- Continuous attributes are typically represented as floating-point variables.



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Countable vs. Measurable

Type	Meaning	Example
Discrete	Finite/countably infinite values	Number of children (0,1,2,...)
Continuous	Infinite possible values (real numbers)	Height, Weight, Time

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Asymmetric Attributes



Only presence (a non-zero attribute value) is regarded as important

Words present in documents
Items present in customer transactions



If we met a friend in the grocery store would we ever say the following?

"I see our purchases are very similar since we didn't buy most of the same things."

We need two asymmetric binary attributes to represent one ordinary binary attribute
Association analysis uses asymmetric attributes
Asymmetric attributes typically arise from objects that are sets

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Some Extensions and Critiques

- Velleman, Paul F., and Leland Wilkinson. "Nominal, ordinal, interval, and ratio typologies are misleading." *The American Statistician* 47, no. 1 (1993): 65-72.
- Mosteller, Frederick, and John W. Tukey. "Data analysis and regression. A second course in statistics." Addison-Wesley Series in Behavioral Science: Quantitative Methods, Reading, Mass.: Addison-Wesley, 1977.
- Chrisman, Nicholas R. "Rethinking levels of measurement for cartography." *Cartography and Geographic Information Systems* 25, no. 4 (1998): 231-242.

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Critiques

- Incomplete
 - Asymmetric binary
 - Cyclical
 - Multivariate
 - Partially ordered
 - Partial membership
 - Relationships between the data
- Real data is approximate and noisy
 - This can complicate recognition of the proper attribute type
 - Treating one attribute type as another may be approximately correct

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Critiques ...

- Not a good guide for statistical analysis
 - May unnecessarily restrict operations and results
 - Statistical analysis is often approximate
 - Thus, for example, using interval analysis for ordinal values may be justified
 - Transformations are common but don't preserve scales
 - Can transform data to a new scale with better statistical properties
 - Many statistical analyses depend only on the distribution

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More Complicated Examples

ID numbers

- Nominal, ordinal, or interval?

Number of cylinders in an automobile engine

- Nominal, ordinal, or ratio?

Biased Scale

- Interval or Ratio

Key Messages for Attribute Types

- The types of operations you choose should be "meaningful" for the type of data you have
 - Distinctness, order, meaningful intervals, and meaningful ratios are only four properties of data
- The data type you see – often numbers or strings – may not capture all the properties or may suggest properties that are not there
- Analysis may depend on these other properties of the data
 - Many statistical analyses depend only on the distribution
- Many times what is meaningful is measured by statistical significance
- But in the end, what is meaningful is measured by the domain

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Types of data sets

Record

- Data Matrix
- Document Data
- Transaction Data

Graph

- World Wide Web
- Molecular Structures

Ordered

- Spatial Data
- Temporal Data
- Sequential Data
- Genetic Sequence Data

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Important Characteristics of Data

Dimensionality (no. of attributes)

- High dimensional data brings a number of challenges

Sparsity

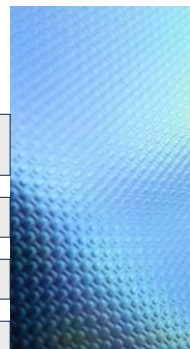
- Only presence counts

Resolution

- Patterns depend on the scale

Size

- Type of analysis may depend on size of data



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Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

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Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute

- Such data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

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Document Data

- Each document becomes a "term" vector
 - Each term is a component (attribute) of the vector
 - The value of each component is the number of times the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

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Transaction Data

- A special type of record data, where
 - Each record (transaction) involves a set of items.
 - For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

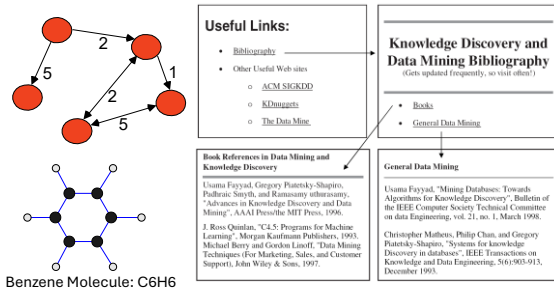
TID	Items
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk



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Graph Data

- Examples: Generic graph, a molecule, and webpages



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Ordered Data

- Sequences of transactions

Items/Events

(A B) (D) (C E)
 (B D) (C) (E)
 (C D) (B) (A E)

An element of the sequence



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Ordered Data

- Genomic sequence data

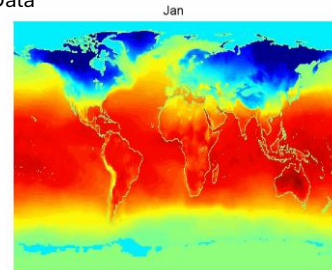
GGTTCCGCCTTCAGCCCCGCGCC
 CGCAGGGCCCCGCCCCGCGCGTC
 GAGAAGGGCCCCGCTGGCGGGCG
 GGGGGAGGCGGGCCGCCGAGC
 CCAACCGAGTCCGACCAGGTGCC
 CCCTCTGCTCGGCCTAGACCTGA
 GCTCATTAGGCGGCAGCGGACAG
 GCCAAGTAGAACACGCGAAGCGC
 TGGGCTGCCTGCTGCGACCAGGG

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Ordered Data

- Spatio-Temporal Data

Average Mon
 Temperature
 land and oce



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Data Quality

- Poor data quality negatively affects many data processing efforts
 "The most important point is that poor data quality is an unfolding disaster.
- Poor data quality costs the typical company at least ten percent (10%) of revenue; twenty percent (20%) is probably a better estimate."

Thomas C. Redman, DM Review, August 2004

- Data mining example: a classification model for detecting people who are loan risks is built using poor data
 - Some credit-worthy candidates are denied loans
 - More loans are given to individuals that default

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Data Quality ...

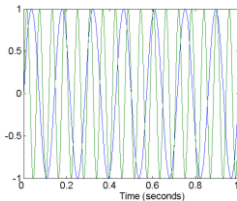
- What kinds of data quality problems?
- How can we detect problems with the data?
- What can we do about these problems?

- Examples of data quality problems:
 - Noise and outliers
 - Missing values
 - Duplicate data
 - Wrong data

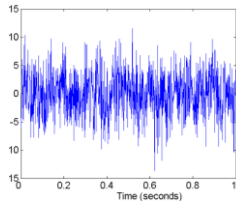
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Noise

- For objects, noise is an extraneous object
- For attributes, noise refers to modification of original values
 - Examples: distortion of a person's voice when talking on a poor phone and "snow" on television screen



Two Sine Waves



Two Sine Waves + Noise

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Outliers

- **Outliers** are data objects with characteristics that are considerably different than most of the other data objects in the data set
- **Case 1:** Outliers are noise that interferes with data analysis
- **Case 2:** Outliers are the goal of our analysis
 - Credit card fraud
 - Intrusion detection
- Causes?

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Missing Values

Reasons for missing values

- Information is not collected (e.g., people decline to give their age and weight)
- Attributes may not be applicable to all cases (e.g., annual income is not applicable to children)

Handling missing values

- Eliminate data objects or variables
- Estimate missing values
 - Example: time series of temperature
 - Example: census results
- Ignore the missing value during analysis

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Missing Values ...

- Missing completely at random (MCAR)
 - Missingness of a value is independent of attributes
 - Fill in values based on the attribute
 - Analysis may be unbiased overall
- Missing at Random (MAR)
 - Missingness is related to other variables
 - Fill in values based on other values
 - Almost always produces a bias in the analysis
- Missing Not at Random (MNAR)
 - Missingness is related to unobserved measurements
 - Informative or non-ignorable missingness
- Not possible to know the situation from the data

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Duplicate Data

- Data set may include data objects that are duplicates, or almost duplicates of one another
 - Major issue when merging data from heterogeneous sources
- Examples:
 - Same person with multiple email addresses
- Data cleaning
 - Process of dealing with duplicate data issues
- When should duplicate data not be removed?

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• 2.2 Statistics of Data – Why?

Understand Data Before Mining

- **Goals:**
 - **Central Tendency** – Where is the middle?
 - **Dispersion** – How spread out?
 - **Visualization** – See patterns, outliers.

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Similarity and Dissimilarity Measures

Similarity measure

- Numerical measure of how alike two data objects are.
- Is higher when objects are more alike.
- Often falls in the range [0,1]

Dissimilarity measure

- Numerical measure of how different two data objects are
- Lower when objects are more alike
- Minimum dissimilarity is often 0
- Upper limit varies

Proximity refers to a similarity or dissimilarity

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2.2.1 Central Tendency – Mean, Median, Mode

The “Average” Family

- Dataset: Salaries (K\$) = [30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110]

Measure	Formula	Calculation	Result
Mean	$\Sigma x_i / N$	$(30 + \dots + 110) / 12$	58
Median	Middle value(s)	$(52 + 56) / 2$	54
Mode	Most frequent	52 & 70	Bimodal

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2.2.2 Dispersion – Range, Quartiles, IQR

- How Spread Out is the Data?

Same dataset: Sorted =

[30,36,47,50,52,52,56,60,63,70,70,110]

Measure	Calculation	Result
Range	Max – Min	$110 - 30 = 80$
Q1 (25th percentile)	Median of first half: $(47+50)/2$	48.5
Q3 (75th percentile)	Median of second half: $(63+70)/2$	66.5
IQR	$Q3 - Q1$	18

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2.2.2 Variance & Standard Deviation

- Measuring Spread Around Mean

Formula: Variance $\sigma^2 = \Sigma(x_i - \text{mean})^2 / N$
Standard Deviation $\sigma = \sqrt{\sigma^2}$

x_i	$x_i - \text{mean}$	$(x_i - \text{mean})^2$
30	-28	784
36	-22	484
...	...	...
110	52	2704
Sum		4550

$$\sigma^2 = 4550 / 12 \approx 379.17$$

$$\sigma \approx \sqrt{379.17} \approx 19.47$$

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2.2.3 Covariance

Do Two Variables Move Together?

Dataset: Stock prices over 5 days

Day	AllElectronics (A)	HighTech (B)
1	6	20
2	5	10
3	4	14
4	3	5
5	2	5

Formula: $\text{Cov}(A,B) = \Sigma[(A_i - \bar{A})(B_i - \bar{B})] / n$

Step-by-Step: $\bar{A} = 4, \bar{B} = 10.8$

$\text{Cov} = [(6-4)(20-10.8) + \dots + (2-4)(5-10.8)] / 5 = 7$

Interpretation: Positive = move together.

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2.2.3 Correlation Coefficient

Standardized Covariance (–1 to +1)

Formula: $r = \text{Cov}(A,B) / (\sigma_A \sigma_B)$

Calculation:

$$\sigma_A = 1.41, \sigma_B = 5.93$$

$$r = 7 / (1.41 \times 5.93) \approx 0.84$$

r Value	Interpretation
+1	Perfect positive linear
0	No linear relationship
–1	Perfect negative linear

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2.2.3 χ^2 Test for Nominal Data

- Are Two Categorical Attributes Related?

Contingency Table: Gender vs. Reading Preference (n=1500)

	Male	Female	Total
Fiction	250	200	450
Non-fiction	50	1000	1050
Total	300	1200	1500

Expected Frequency Calculation

$$E_{ij} = (\text{RowTotal} \times \text{ColTotal}) / N$$

$$E_{11} = (450 \times 300) / 1500 = 90$$

$$\chi^2 \text{ Formula: } \chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}$$

$$= 507.93$$

Result: Compare to χ^2 table (df=1).
Significant $\Rightarrow$ correlated.

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2.2.4 Graphic Displays – Quantile Plot

Definition: A quantile plot displays ordered data values (sorted ascending) against their corresponding cumulative probabilities.

Formula:

For each sorted observation x_i ($i = 1$ to N):

$$f_i = (i - 0.5) / N \text{ Where:}$$

- f_i = proportion of data below x_i
- N = total number of observation

Example Dataset: Unit Prices (\$)

Data from Table 2.3 (simplified):

* [40,43,47,49,50,52,55,60,65,70,72,75,80,85,90,95,100,110,120][40,43,47,49,50,52,55,60,65,70,72,75,80,85,90,95,100,110,120] **N = 19 observations**

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Step 1: Sort data (already sorted)

Step 2: Compute f_i for each observation:

i	x_i (\$)	$f_i = (i - 0.5) / 19$	Interpretation
1	40	(1-0.5)/19 = 0.0263	2.63% of items $\leq$ \$40
2	43	(2-0.5)/19 = 0.0789	7.89% of items $\leq$ \$43
3	47	(3-0.5)/19 = 0.1316	13.16% of items $\leq$ \$47
4	49	(4-0.5)/19 = 0.1842	18.42% of items $\leq$ \$49
5	50	(5-0.5)/19 = 0.2368	23.68% of items $\leq$ \$50
6	52	0.2895	28.95% of items $\leq$ \$52
7	55	0.3421	34.21% of items $\leq$ \$55
8	60	0.3947	39.47% of items $\leq$ \$60
9	65	0.4474	44.74% of items $\leq$ \$65
10	70	0.5000	50% of items $\leq$ \$70 (Median)
11	72	0.5526	55.26% of items $\leq$ \$72
12	75	0.6053	60.53% of items $\leq$ \$75
13	80	0.6579	65.79% of items $\leq$ \$80
14	85	0.7105	71.05% of items $\leq$ \$85
15	90	0.7632	76.32% of items $\leq$ \$90
16	95	0.8158	81.58% of items $\leq$ \$95
17	100	0.8684	86.84% of items $\leq$ \$100
18	110	0.9211	92.11% of items $\leq$ \$110
19	120	(19-0.5)/19 = 0.9737	97.37% of items $\leq$ \$120

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Finding Q1 and Q3:

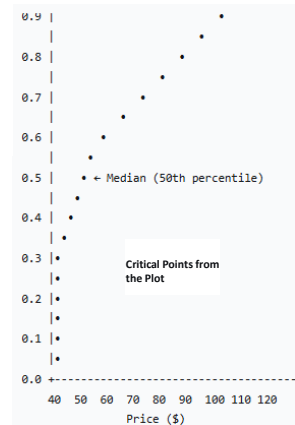
Since we don't have exact $f_i = 0.25$ or 0.75, we interpolate:

• For $f_i = 0.25$: between $x=50$ ($f=0.2368$) and $x=52$ ($f=0.2895$)

• Linear interpolation: $Q1 \approx \$51.60$

• Similarly, $Q3 \approx \$94.80$

Point	Price (\$)	f_i	Interpretation
Q1 (25th percentile)	~52	0.25	25% of items cost $\leq$ \$52
Median (50th percentile)	70	0.50	Half of items cost $\leq$ \$70
Q3 (75th percentile)	~95	0.75	75% of items cost $\leq$ \$95

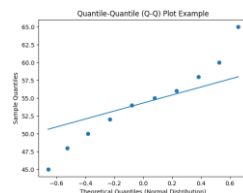


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Quantile-Quantile (Q-Q) Plot

A **Quantile-Quantile (Q-Q) plot** compares the **quantiles of a dataset** with the **quantiles of a theoretical distribution** (usually Normal) to check:

- Whether the data follows that distribution
- Presence of skewness
- Presence of outliers
- ★ If points lie on a straight line $\rightarrow$ distributions match



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Dataset Comparison: Two Store Branches

Branch 1 (10 items): [62, 65, 68, 70, 72, 75, 78, 80, 82, 85]

Branch 2 (12 items): [60, 63, 65, 67, 69, 70, 72, 74, 76, 78, 80, 82]

Step 1: Calculate quantiles for Each Dataset

Formula: $F_i = i - 0.5 / n$

Branch 1 ($n_1=10$):

i	Price	f_i (quantiles)	Interpretation
1	62	(1-0.5)/10 = 0.05	5th percentile
2	65	0.15	15th percentile
3	68	0.25	Q1: 25th percentile
4	70	0.35	
5	72	0.45	
6	75	0.55	Median: 50th percentile
7	78	0.65	
8	80	0.75	Q3: 75th percentile
9	82	0.85	
10	85	0.95	95th percentile

Branch 2 ($n_2=12$):

i	Price	f_i (quantile)
1	60	0.0417
2	63	0.125
3	65	0.2083
4	67	0.2917
5	69	0.375
6	70	0.4583
7	72	0.5417
8	74	0.625
9	76	0.7083
10	78	0.7917
11	80	0.875
12	82	0.9583

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Step 2: Match Quantiles Between Branches

Since $n_1 \neq n_2$, we interpolate Branch 2 values at Branch 1's percentiles:

Interpolation formula:

$$\text{Value} = \text{lower} + (f - \text{lower}) \times ((\text{upper} - \text{lower}) / (f_{\text{upper}} - f_{\text{lower}}))$$

Example: $f=0.25$ (25th percentile/quantile)

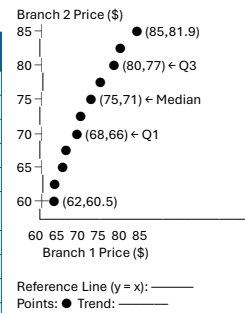
• Branch 1: 68 (directly)

• Branch 2: Between $i=3$ ($f=0.2083$, value=65) and $i=4$ ($f=0.2917$, value=67)

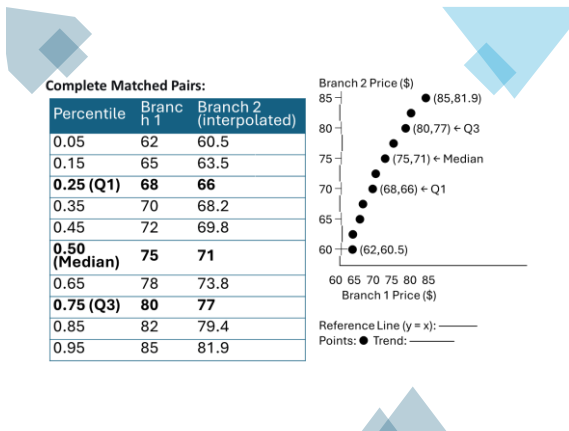
$$\begin{aligned} B2_{0.25} &= 65 + (0.25 - 0.2083) \times (67 - 65) / (0.2917 - 0.2083) \\ &= 65 + 0.0417 \times 2 / 0.0834 = 65 + 1 = 66 \end{aligned}$$

Complete Matched Pairs:

Percentile	Branch 1	Branch 2 (interpolated)
0.05	62	60.5
0.15	65	63.5
0.25 (Q1)	68	66
0.35	70	68.2
0.45	72	69.8
0.50 (Median)	75	71
0.65	78	73.8
0.75 (Q3)	80	77
0.85	82	79.4
0.95	85	81.9

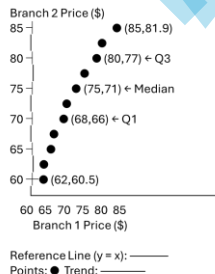


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Complete Matched Pairs:

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0.75 (Q3)	80	77
0.85	82	79.4
0.95	85	81.9



Interpretation Pattern Analysis:

Below line ($y=x$): Branch 2 prices < Branch 1 at same percentile

Consistent gap: ~\$4-5 difference across all percentiles

No crossing: Same relative ranking maintained

Branch 1: Mean = 73.7, SD = 6.7

Branch 2: Mean = 70.2, SD = 6.3

• **Systematic shift:** Branch 1 prices consistently higher

• **Similar spread:** Comparable standard deviations

• **No shape distortion:** Parallel to $y=x$ line indicates same distribution shape

Quartile Comparison Table

Quartile	Branch 1	Branch 2	Difference
Q1 (25th)	\$68	\$66	-\$2
Median (50th)	\$75	\$71	-\$4
Q3 (75th)	\$80	\$77	-\$3
IQR	\$12	\$11	-\$1

Quick interpretation points

Pattern:	Meaning:
• Points on $y=x$ line	→ Identical distributions
• Points below line	→ Branch 1 > Branch 2
• Points above line	→ Branch 2 > Branch 1
• Curved pattern	→ Different distribution shapes
• Scattered points	→ No consistent relationship

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2.2.4 Histograms & Scatter Plots

• See Patterns & Relationships

• *Histograms are like bar charts for introverts—they don't like gaps.*

Graph Type	Purpose	Example
Histogram	Frequency distribution	Price ranges
Scatter Plot	Bivariate relationship	Income vs. Spending
Boxplot	5-number summary	Salary outliers

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Interpretation Pattern Analysis:

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Consistent gap: ~\$4-5 difference across all percentiles

No crossing: Same relative ranking maintained

Branch 1: Mean = 73.7, SD = 6.7

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Q3 (75th)	\$80	\$77	-\$3
IQR	\$12	\$11	-\$1

Quick interpretation points

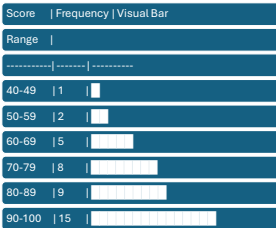
Pattern:	Meaning:
• Points on $y=x$ line	→ Identical distributions
• Points below line	→ Branch 1 > Branch 2
• Points above line	→ Branch 2 > Branch 1
• Curved pattern	→ Different distribution shapes
• Scattered points	→ No consistent relationship

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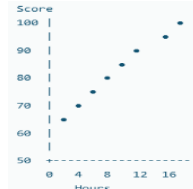
60

Histograms & Scatter Plots

[45, 55, 60, 62, 65, 65, 68, 70, 72, 72, 75, 75, 75, 78, 78, 80, 82, 82, 85, 85, 85, 88, 88, 90, 90, 92, 92, 92, 95, 95, 95, 95, 98, 98, 98, 98, 98, 99, 99, 100]



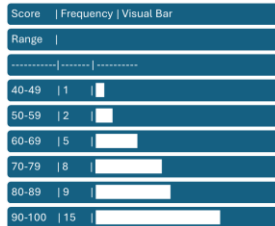
Student	Hours Studied	Exam Score
1	2	55
2	4	65
3	6	75
4	8	85
5	10	88
6	12	92
7	14	94
8	16	96



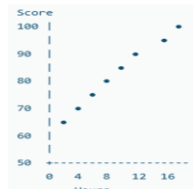
61

Histograms & Scatter Plots

[45, 55, 60, 62, 65, 65, 68, 70, 72, 72, 75, 75, 75, 78, 78, 80, 82, 82, 85, 85, 85, 88, 88, 90, 90, 92, 92, 92, 95, 95, 95, 95, 98, 98, 98, 98, 98, 99, 99, 100]



Student	Hours Studied	Exam Score
1	2	55
2	4	65
3	6	75
4	8	85
5	10	88
6	12	92
7	14	94
8	16	96



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2.3 Similarity & Distance Measures – Why?

The Glue of Clustering & Classification

Applications:

- Clustering (group similar customers)
- Nearest-neighbor classification
- Outlier detection (far from others)

• **Quote:** “In data mining, birds of a feather flock together—if we measure feathers right.”

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2.3.1 Data Matrix vs. Dissimilarity Matrix

Two Ways to Structure Data

Data Matrix (n × p):

Object	Attr1	Attr2	Attr3
1	5.1	3.5	1.4
2	4.9	3.0	1.4
...	...	...	...

Dissimilarity Matrix (n × n):

	1	2	3
1	0	0.2	0.7
2	0.2	0	0.5
3	0.7	0.5	0

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Distance Measures

Remember K-Nearest Neighbor are determined on the bases of some kind of “distance” between points.

Two major classes of distance measure:

- **Euclidean**: based on position of points in some k -dimensional space.
- **Noneuclidean**: not related to position or space.

Applying a distance measure largely depends on the type of input data

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Axioms of a Distance Measure

- d is a distance measure if it is a function from pairs of points to reals such that:
 - **Non-negativity**: $d(i,j) \geq 0$.
 - **Identity of indiscernibles**: $d(i,j) = 0$: The distance of an object to itself is 0
 - **Symmetry**: $d(i,j) = d(j,i)$.
 - **Triangle inequality**: $d(i,j) \leq d(i,k) + d(k,j)$
- Any measure that satisfies these conditions is known as metric.

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2.3.2 Proximity for Nominal Attributes

- The dissimilarity between two objects i and j having nominal attributes can be computed based on the ratio of mismatches
- $$d(i,j) = \frac{p-m}{p} \quad \text{sim}(i,j) = 1 - d(i,j) = \frac{m}{p}$$
- M : number of matches
 p : number of attributes.

Example: Two patients:

Attr	Pat1	Pat2	Match?
Gender	M	M	Yes
Blood Type	A	B	No
Smoker	Y	N	No

$p=3, m=1 \Rightarrow d = (3-1)/3 = 0.67$
Similarity = 1 - d = 0.33

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2.3.3 Binary Proximity – Jaccard Coefficient

Jaccard Similarity:
 $\text{sim} = q / (q + r + s)$
Ignore negative matches (t)
Example:
Patients with symptoms:
 $q=2$ (both have fever & cough)
 $r=1$ (Pat1 has headache, Pat2 not)
 $s=1$ (Pat2 has rash, Pat1 not)
 $\text{sim} = 2/(2+1+1) = 0.5$

Asymmetric Binary
Similarity
Contingency Table:

	j=1	j=0	Sum
i=1	q	r	q+r
i=0	s	t	s+t
Sum	q+s	r+t	p

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2.3.4 Numeric Distance – Euclidean & Manhattan

Title: Straight Line vs. City Blocks
Points: A=(1,2), B=(3,5)

Distance	Formula	Calculation	Result
Euclidean	$\sqrt{\sum (x_i - y_i)^2}$	$\sqrt{((1-3)^2 + (2-5)^2)}$	$\sqrt{13} \approx 3.61$
Manhattan	$\sum x_i - y_i $	$ 1-3 + 2-5 $	=5
Supremum	$\max(x_i - y_i)$	$\max(2,3)$	=3

Image: Grid showing both distances
Joke: Euclidean is for crows, Manhattan is for taxi drivers.

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Similarity / Dissimilarity for Simple Attributes

- The following table shows the similarity and dissimilarity between two objects, x and y , with respect to a single, simple attribute.

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$
Ordinal	$d = x - y / (n - 1)$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - d$
Interval or Ratio	$d = x - y $	$s = -d, s = \frac{d}{1+d}, s = e^{-d}, s = 1 - \frac{d - \min d}{\max d - \min d}$

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Some Euclidean Distances

L_2 norm (also common or Euclidean distance):

- The most common notion of “distance”.

$$d(i,j) = \sqrt{(|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \dots + |x_{ip} - x_{jp}|^2)}$$

L_1 norm (also called Manhattan distance)

- distance if you had to travel along coordinates only.

$$d(i,j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|$$

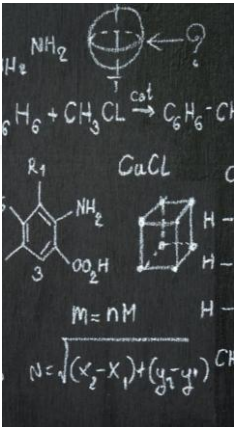
71

Euclidean Distance

- where n is the number of dimensions (attributes) and x_k and y_k are, respectively, the k^{th} attributes (components) or data objects $\mathbf{x}$ and $\mathbf{y}$.

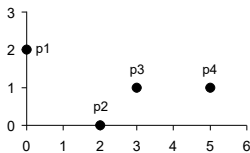
$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

- Standardization is necessary, if scales differ.



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Euclidean Distance



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1



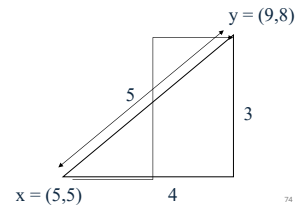
	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

• Distance Matrix

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Examples L_1 and L_2 norms

- L_1 -norm: $\bullet \text{dist}(x,y) = |5-9| + |5-8| = 4+3 = 7$
- L_2 -norm: $\bullet \text{dist}(x,y) = \sqrt{(|5-9|^2 + |5-8|^2)} = \sqrt{4^2 + 3^2} = 5$



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Minkowski Distance

- Minkowski Distance is a generalization of Euclidean Distance

$$d(x, y) = \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r}$$

Where r is a parameter, n is the number of dimensions (attributes) and x_k and y_k are, respectively, the k^{th} attributes (components) or data objects x and y .

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Minkowski Distance: Examples

- $r = 1$. City block (Manhattan, taxicab, L_1 norm) distance.
 - A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors
- $r = 2$. Euclidean distance
- $r \rightarrow \infty$. "supremum" ($L_{\max}$ norm, L_{∞} norm) distance.
 - This is the maximum difference between any component of the vectors
- Do not confuse r with n , i.e., all these distances are defined for all numbers of dimensions.

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Minkowski Distance

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_{∞}	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

• Distance Matrix

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Another Euclidean Distance

- L_{∞} norm (Supremum distance-Uniform distance): $d(x,y)$ = the maximum of the differences between x and y in any dimension.

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{1/h} = \max_f |x_{if} - x_{jf}|$$

- Example: Let we have following two points

$$x = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \\ 7 \end{bmatrix} \quad y = \begin{bmatrix} 3 \\ 8 \\ 6 \\ 2 \\ 5 \end{bmatrix}$$

- and Find L_1 , L_2 and L_{∞}

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Non-Euclidean Distances

- Proximity Measure** for Nominal Attributes
- Jaccard measure** for binary vectors
- Distance measure** for Ordinal variables
- Edit distance** = number of inserts and deletes to change one string into another.
- Cosine measure** = angle between vectors from the origin to the points in question.



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2.3.5 Ordinal Proximity – Step-by-Step

Convert to Numeric, Then Distance

Example: Education level {HS=1, Bachelor=2, Master=3, PhD=4}

Steps:

1. Replace with rank r_i
2. Normalize: $z_i = (r_i - 1)/(M - 1)$ where $M=4$
 1. HS: $z = (1 - 1)/3 = 0$
 2. Bachelor: $z = (2 - 1)/3 = 0.33$
 3. Master: $z = 0.67$
 4. PhD: $z = 1$
3. Use Euclidean distance on z_i

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2.3.6 Mixed Attributes – Weighted Average

Combining Different Attribute Types

Formula:

$$d(i,j) = \sum \delta_{ij}^{(n)} d_{ij}^{(n)} / \sum \delta_{ij}^{(n)}$$

Example: Customer data:

$$\delta = 1 \text{ for all.}$$

$$d = (0.1 + 1 + 0.33)/3 \approx 0.48$$

Attr	Type	Value i	Value j	$d_{ij}^{(n)}$	
Age	Numeric	25	30	25-30	/ (max-min) = 5/50=0.1
Gender	Binary (sym)	M	F	1	
Education	Ordinal	Bachelor	Master	0.33	

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Two Job Applicants

Attribute	Type	Applicant A	Applicant B	Notes
Age	Numeric	25	35	Range in dataset: 20-60
Degree Level	Ordinal	Bachelor	PhD	Ranks: HS=1, Bachelor=2, Master=3, PhD=4
Programming Test Passed	Asymmetric Binary	Yes (1)	No (0)	1=Passed, 0=Failed
Has Reference Letter	Symmetric Binary	Yes (1)	Yes (1)	1=Yes, 0=No
Years Experience	Numeric	2	8	Range in dataset: 0-15

Missing: None (all $\delta=1$)

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Missing: None (all $\delta=1$)

Individual Attribute Dissimilarities

1. Age (Numeric) $d_{Age} = (|25 - 35|) / (max - min) = 10/40 = 0.25$

2. Degree Level (Ordinal)

Step A: Convert to ranks

Bachelor = 2, PhD = 4,

$r_A=2, r_B=4$

Step B: Normalize to [0,1]

$z = (r - 1)/(M - 1)$, $M=4$

$z_A = (2 - 1)/3 = 0.333$, $z_B = (4 - 1)/3 = 1$

$d_{Degree} = |0.333 - 1| = 0.667$

3. Programming Test (Asymmetric Binary)

Contingency table: $q=0, r=1, s=0$

$d_{Prog} = (r+s)/(q+r+s) = (1+0)/(0+1+0) = 1$

5. Years Experience (Numeric)

$d_{Exp} = |2 - 8| / (15 - 0) = 6/15 = 0.4$

4. Reference Letter (Symmetric Binary)

Both have it $\rightarrow d = 0$

Attribute	Type	Applicant A	Applicant B	Notes
Age	Numeric	25	35	Range in dataset: 20-60
Degree Level	Ordinal	Bachelor	PhD	Ranks: HS=1, Bachelor=2, Master=3, PhD=4
Programming Test Passed	Asymmetric Binary	Yes (1)	No (0)	1=Passed, 0=Failed
Has Reference Letter	Symmetric Binary	Yes (1)	Yes (1)	1=Yes, 0=No
Years Experience	Numeric	2	8	Range in dataset: 0-15

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Combined Dissimilarity Table

Attribute	Type	$d(i,j)$	Weight (δ)	$\delta \times d(i,j)$
Age	Numeric	0.25	1	0.25
Degree	Ordinal	0.667	1	0.667
Programming Test	Asym. Binary	1	1	1
Reference Letter	Sym. Binary	0	1	0
Years Experience	Numeric	0.4	1	1
Totals			$\sum \delta = 5$	$\sum [\delta \times d] = 2.317$

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Final Calculation

$$d(A, B) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}} = \frac{2.317}{5} = 0.4634$$

Interpretation & Business Insight

Similarity Score:

$$\text{sim}(A, B) = 1 - d(A, B) = 1 - 0.463 = 0.537 \text{ (53.7\% similar)}$$

Hiring Decision Analysis:

- **Major differences:**
 - Programming test (failed vs passed) + Education level
- **Similarities:** Both have reference letters
- **Moderate differences:** Age (10 years) and Experience (6 years)
- **Cluster Placement:** In HR analytics, these two would likely be in **different clusters** for:
 - Technical roles (programming test crucial)
 - Seniority levels (PhD vs Bachelor + experience difference)

Attribute	Type	Applicant A	Applicant B	Notes
Age	Numeric	25	35	Range in dataset: 20-60
Degree Level	Ordinal	Bachelor	PhD	Ranks: HS=1, Bachelor=2, Master=3, PhD=4
Programming Test Passed	Asymmetric Binary	Yes (1)	No (0)	1=Passed, 0=Failed
Has Reference Letter	Symmetric Binary	Yes (1)	Yes (1)	1=Yes, 0=No
Years Experience	Numeric	2	8	Range in dataset: 0-15

2.3.7 Cosine Similarity – Document Similarity

Measure Angle Between Vectors

Formula:

$$\cos(\theta) = A \cdot B / (||A|| \cdot ||B||)$$

Example:

$$\text{Doc1} = [5, 0, 3, 0, 2, 0, 0, 2, 0, 0]$$

$$\text{Doc2} = [3, 0, 2, 0, 1, 1, 0, 1, 0, 1]$$

Calculation:

$$\text{Dot product} = 5 \times 3 + 0 \times 0 + 3 \times 2 + \dots = 25$$

$$||\text{Doc1}|| = \sqrt{5^2 + 0^2 + 3^2 + \dots} = \sqrt{42} \approx 6.48$$

$$||\text{Doc2}|| = \sqrt{3^2 + 0^2 + 2^2 + \dots} = \sqrt{17} \approx 4.12$$

$$\cos = 25 / (6.48 \times 4.12) \approx 0.94$$

Interpretation: Near 1 $\Rightarrow$ very similar topics.



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2.3.8 KL Divergence – Comparing Distributions

Title: Information Lost When Approximating

Formula:

$$D_{KL}(P||Q) = \sum P(x) \log(P(x)/Q(x))$$

Example:

$$P = [0.6, 0.4] \text{ (true distribution)}$$

$$Q = [0.5, 0.5] \text{ (approximation)}$$

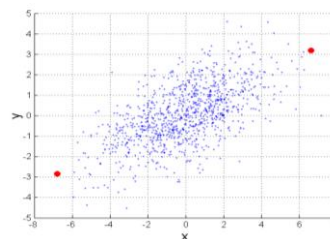
$$D_{KL} = 0.6 \log(0.6/0.5) + 0.4 \log(0.4/0.5) \approx 0.0204$$

Properties:

- Not symmetric
- $D_{KL} \geq 0$
- 0 if $P=Q$

Mahalanobis Distance

$$\text{mahalanobis}(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})^T \Sigma^{-1} (\mathbf{x} - \mathbf{y})$$

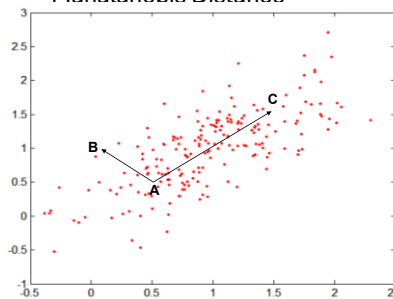


Σ is the covariance matrix

For red points, the Euclidean distance is 14.7, Mahalanobis distance is 6.

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Mahalanobis Distance



Covariance Matrix:

$$\Sigma = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$$

$$\mathbf{A}: (0.5, 0.5)$$

$$\mathbf{B}: (0, 1)$$

$$\mathbf{C}: (1.5, 1.5)$$

$$\text{Mahal}(\mathbf{A}, \mathbf{B}) = 5$$

$$\text{Mahal}(\mathbf{A}, \mathbf{C}) = 4$$

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Common Properties of a Distance

- Distances, such as the Euclidean distance, have some well known properties.

1. $d(\mathbf{x}, \mathbf{y}) \geq 0$ for all $\mathbf{x}$ and $\mathbf{y}$ and $d(\mathbf{x}, \mathbf{y}) = 0$ only if $\mathbf{x} = \mathbf{y}$. (Positive definiteness)
2. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}$ and $\mathbf{y}$. (Symmetry)
3. $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$ for all points $\mathbf{x}, \mathbf{y}$, and $\mathbf{z}$. (Triangle Inequality)

where $d(\mathbf{x}, \mathbf{y})$ is the distance (dissimilarity) between points (data objects), $\mathbf{x}$ and $\mathbf{y}$.

- A distance that satisfies these properties is a **metric**

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Common Properties of a Similarity

- Similarities, also have some well known properties.

1. $s(\mathbf{x}, \mathbf{y}) = 1$ (or maximum similarity) only if $\mathbf{x} = \mathbf{y}$.
2. $s(\mathbf{x}, \mathbf{y}) = s(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}$ and $\mathbf{y}$. (Symmetry)

where $s(\mathbf{x}, \mathbf{y})$ is the similarity between points (data objects), $\mathbf{x}$ and $\mathbf{y}$.

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SMC versus Jaccard: Example

$\mathbf{x} = 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$
 $\mathbf{y} = 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 1$

$f_{01} = 2$ (the number of attributes where p was 0 and q was 1)
 $f_{10} = 1$ (the number of attributes where p was 1 and q was 0)
 $f_{00} = 7$ (the number of attributes where p was 0 and q was 0)
 $f_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$$\text{SMC} = (f_{11} + f_{00}) / (f_{01} + f_{10} + f_{11} + f_{00}) \\ = (0 + 7) / (2 + 1 + 0 + 7) = 0.7$$

$$J = (f_{11}) / (f_{01} + f_{10} + f_{11}) = 0 / (2 + 1 + 0) = 0$$

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Extended Jaccard Coefficient (Tanimoto)

- Variation of Jaccard for continuous or count attributes
 - Reduces to Jaccard for binary attributes

$$EJ(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \mathbf{x} \cdot \mathbf{y}}$$

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Similarity Between Binary Vectors

- Common situation is that objects, p and q , have only binary attributes
- Compute similarities using the following quantities
 - f_{01} = the number of attributes where p was 0 and q was 1
 - f_{10} = the number of attributes where p was 1 and q was 0
 - f_{00} = the number of attributes where p was 0 and q was 0
 - f_{11} = the number of attributes where p was 1 and q was 1
- Simple Matching and Jaccard Coefficients
 - $\text{SMC} = \text{number of matches} / \text{number of attributes}$
 $= (f_{11} + f_{00}) / (f_{01} + f_{10} + f_{11} + f_{00})$
 - $J = \text{number of 11 matches} / \text{number of non-zero attributes}$
 $= (f_{11}) / (f_{01} + f_{10} + f_{11})$

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Cosine Similarity

- If $\mathbf{d}_1$ and $\mathbf{d}_2$ are two document vectors, then
 - $\cos(\mathbf{d}_1, \mathbf{d}_2) = \langle \mathbf{d}_1, \mathbf{d}_2 \rangle / \|\mathbf{d}_1\| \|\mathbf{d}_2\|$,
 where $\langle \mathbf{d}_1, \mathbf{d}_2 \rangle$ indicates inner product or vector dot product of vectors, $\mathbf{d}_1$ and $\mathbf{d}_2$, and $\|\mathbf{d}\|$ is the length of vector $\mathbf{d}$.

- Example:

$$\mathbf{d}_1 = 3\ 2\ 0\ 5\ 0\ 0\ 0\ 2\ 0\ 0 \\ \mathbf{d}_2 = 1\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 2 \\ \langle \mathbf{d}_1, \mathbf{d}_2 \rangle = 3*1 + 2*0 + 0*0 + 5*0 + 0*0 + 0*0 + 0*2 + 0*1 + 0*0 + 0*2 = 5 \\ \|\mathbf{d}_1\| = (3^2 + 2^2 + 0^2 + 5^2 + 0^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2)^{0.5} = (42)^{0.5} = 6.481 \\ \|\mathbf{d}_2\| = (1^2 + 0^2 + 0^2 + 0^2 + 0^2 + 0^2 + 1^2 + 0^2 + 2^2)^{0.5} = (6)^{0.5} = 2.449 \\ \cos(\mathbf{d}_1, \mathbf{d}_2) = 0.3150$$

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Correlation measures the linear relationship between objects

$$\text{corr}(\mathbf{x}, \mathbf{y}) = \frac{\text{covariance}(\mathbf{x}, \mathbf{y})}{\text{standard_deviation}(\mathbf{x}) * \text{standard_deviation}(\mathbf{y})} = \frac{s_{xy}}{s_x s_y} \quad (2.11)$$

where we are using the following standard statistical notation and definitions

$$\text{covariance}(\mathbf{x}, \mathbf{y}) = s_{xy} = \frac{1}{n-1} \sum_{k=1}^n (x_k - \bar{x})(y_k - \bar{y}) \quad (2.12)$$

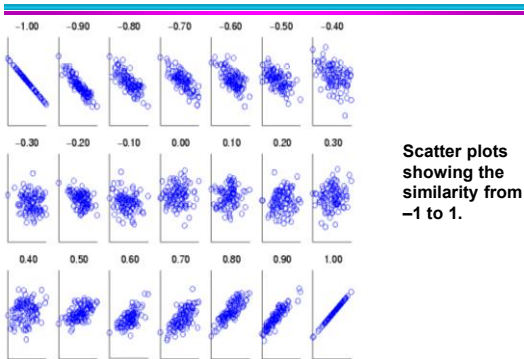
$$\text{standard_deviation}(\mathbf{x}) = s_x = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (x_k - \bar{x})^2} \\ \text{standard_deviation}(\mathbf{y}) = s_y = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (y_k - \bar{y})^2}$$

$$\bar{x} = \frac{1}{n} \sum_{k=1}^n x_k \text{ is the mean of } \mathbf{x}$$

$$\bar{y} = \frac{1}{n} \sum_{k=1}^n y_k \text{ is the mean of } \mathbf{y}$$

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Visually Evaluating Correlation



Scatter plots showing the similarity from -1 to 1.

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Drawback of Correlation

- $x = (-3, -2, -1, 0, 1, 2, 3)$
- $y = (9, 4, 1, 0, 1, 4, 9)$

$$y_i = x_i^2$$

- $\text{mean}(x) = 0, \text{mean}(y) = 4$
- $\text{std}(x) = 2.16, \text{std}(y) = 3.74$

$$\text{corr} = \frac{(-3)(5) + (-2)(0) + (-1)(-3) + (0)(-4) + (1)(-3) + (2)(0) + 3(5)}{6 * 2.16 * 3.74} = 0$$

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Comparison of Proximity Measures

- Domain of application
 - Similarity measures tend to be specific to the type of attribute and data
 - Record data, images, graphs, sequences, 3D-protein structure, etc. tend to have different measures
- However, one can talk about various properties that you would like a proximity measure to have
 - Symmetry is a common one
 - Tolerance to noise and outliers is another
 - Ability to find more types of patterns?
 - Many others possible
- The measure must be applicable to the data and produce results that agree with domain knowledge

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Information Based Measures

- Information theory is a well-developed and fundamental discipline with broad applications
- Some similarity measures are based on information theory
 - Mutual information in various versions
 - Maximal Information Coefficient (MIC) and related measures
 - General and can handle non-linear relationships
 - Can be complicated and time intensive to compute

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Information and Probability

- Information relates to possible outcomes of an event
 - transmission of a message, flip of a coin, a piece of data
- The more certain an outcome, the less information that it contains and vice-versa
 - For example, if a coin has two heads, then an outcome of heads provides no information
 - More quantitatively, the information is related to the probability of an outcome
 - The smaller the probability of an outcome, the more information it provides and vice-versa
- Entropy is the commonly used measure



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Entropy

- For
 - a variable (event), X ,
 - with n possible values (outcomes), $x_1, x_2, \dots, x_n$
 - each outcome having probability, $p_1, p_2, \dots, p_n$
 - the entropy of X , $H(X)$, is given by
- $$H(X) = - \sum_{i=1}^n p_i \log_2 p_i$$
- Entropy is between 0 and $\log_2 n$ and is measured in bits
 - Thus, entropy is a measure of how many bits it takes to represent an observation of X on average

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Entropy Examples

- For a coin with probability p of heads and probability $q = 1 - p$ of tails
$$H = -p \log_2 p - q \log_2 q$$
 - For $p = 0.5, q = 0.5$ (fair coin) $H = 1$
 - For $p = 1$ or $q = 1, H = 0$
- What is the entropy of a fair four-sided die?

Entropy for Sample Data: Example

Hair Color	Count	p	$-p \log_2 p$
Black	75	0.75	0.3113
Brown	15	0.15	0.4105
Blond	5	0.05	0.2161
Red	0	0.00	0
Other	5	0.05	0.2161
Total	100	1.0	1.1540

Maximum entropy is $\log_2 5 = 2.3219$

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Entropy for Sample Data

- Suppose we have
 - a number of observations (m) of some attribute, X , e.g., the hair color of students in the class,
 - where there are n different possible values
 - And the number of observation in the i^{th} category is m_i
 - Then, for this sample
$$H(X) = - \sum_{i=1}^n \frac{m_i}{m} \log_2 \frac{m_i}{m}$$
- For continuous data, the calculation is harder

Mutual Information

- Information one variable provides about another
Formally, $I(X, Y) = H(X) + H(Y) - H(X, Y)$, where
 $H(X, Y)$ is the joint entropy of X and Y ,
$$H(X, Y) = - \sum_i \sum_j p_{ij} \log_2 p_{ij}$$

Where p_{ij} is the probability that the i^{th} value of X and the j^{th} value of Y occur together
- For discrete variables, this is easy to compute
- Maximum mutual information for discrete variables is $\log_2(\min(n_X, n_Y))$, where n_X (n_Y) is the number of values of X (Y)

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Mutual Information Example

Student Status	Count	p	$-p \log_2 p$
Undergrad	45	0.45	0.5184
Grad	55	0.55	0.4744
Total	100	1.00	0.9928

Grade	Count	p	$-p \log_2 p$
A	35	0.35	0.5301
B	50	0.50	0.5000
C	15	0.15	0.4105
Total	100	1.00	1.4406

Student Status	Grade	Count	p	$-p \log_2 p$
Undergrad	A	5	0.05	0.2161
Undergrad	B	30	0.30	0.5211
Undergrad	C	10	0.10	0.3322
Grad	A	30	0.30	0.5211
Grad	B	20	0.20	0.4644
Grad	C	5	0.05	0.2161
Total		100	1.00	2.2710

Mutual information of Student Status and Grade = $0.9928 + 1.4406 - 2.2710 = 0.1624$

Maximal Information Coefficient

- Reshef, David N., Yakir A. Reshef, Hilary K. Finucane, Sharon R. Grossman, Gilean McVean, Peter J. Turnbaugh, Eric S. Lander, Michael Mitzenmacher, and Pardis C. Sabeti. "Detecting novel associations in large data sets." *science* 334, no. 6062 (2011): 1518-1524.
- Applies mutual information to two continuous variables
- Consider the possible binnings of the variables into discrete categories
 - $n_X \times n_Y \leq N^{0.6}$ where
 - n_X is the number of values of X
 - n_Y is the number of values of Y
 - N is the number of samples (observations, data objects)
- Compute the mutual information
 - Normalized by $\log_2(\min(n_X, n_Y))$
- Take the highest value

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General Approach for Combining Similarities

- Sometimes attributes are of many different types, but an overall similarity is needed.
- For the k^{th} attribute, compute a similarity, $s_k(\mathbf{x}, \mathbf{y})$, in the range $[0, 1]$.
 - Define an indicator variable, δ_k , for the k^{th} attribute as follows:
 - $\delta_k = 0$ if the k^{th} attribute is an asymmetric attribute and both objects have a value of 0, or if one of the objects has a missing value for the k^{th} attribute
 - $\delta_k = 1$ otherwise
 - Compute

$$\text{similarity}(\mathbf{x}, \mathbf{y}) = \frac{\sum_{k=1}^n \delta_k s_k(\mathbf{x}, \mathbf{y})}{\sum_{k=1}^n \delta_k}$$

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Density

- Measures the degree to which data objects are close to each other in a specified area
- The notion of density is closely related to that of proximity
- Concept of density is typically used for clustering and anomaly detection
- Examples:
 - Euclidean density
 - Euclidean density = number of points per unit volume
 - Probability density
 - Estimate what the distribution of the data looks like
 - Graph-based density
 - Connectivity

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Euclidean Density: Center-Based

- Euclidean density is the number of points within a specified radius of the point

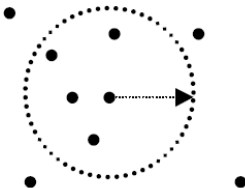


Illustration of center-based density.

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Using Weights to Combine Similarities

- May not want to treat all attributes the same.
 - Use non-negative weights ω_k

$$\text{similarity}(\mathbf{x}, \mathbf{y}) = \frac{\sum_{k=1}^n \omega_k \delta_k s_k(\mathbf{x}, \mathbf{y})}{\sum_{k=1}^n \omega_k \delta_k}$$

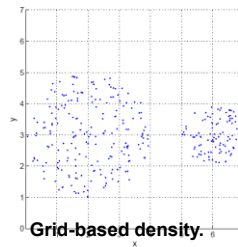
- Can also define a weighted form of distance

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n w_k |x_k - y_k|^r \right)^{1/r}$$

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Euclidean Density: Grid-based Approach

- Simplest approach is to divide region into a number of rectangular cells of equal volume and define density as # of points the cell contains



Grid-based density.

0	0	0	0	0	0	0
0	0	0	0	0	0	0
4	17	18	6	0	0	0
14	14	13	13	0	18	27
11	18	10	21	0	24	31
3	20	14	4	0	0	0
0	0	0	0	0	0	0

Counts for each cell.

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Data Preprocessing

- Aggregation
- Sampling
- Dimensionality Reduction
- Feature subset selection
- Feature creation
- Discretization and Binarization
- Attribute Transformation

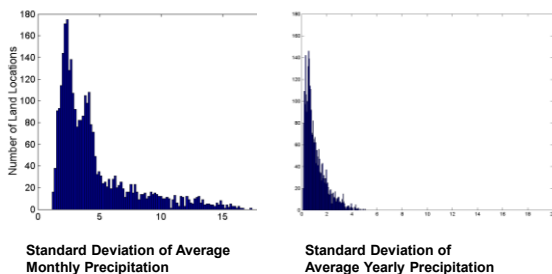
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Aggregation

- Combining two or more attributes (or objects) into a single attribute (or object)
- Purpose
 - Data reduction
 - Reduce the number of attributes or objects
 - Change of scale
 - Cities aggregated into regions, states, countries, etc.
 - Days aggregated into weeks, months, or years
 - More "stable" data
 - Aggregated data tends to have less variability

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Example: Precipitation in Australia ... Variation of Precipitation in Australia



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Sampling ...

- The key principle for effective sampling is the following:
 - Using a sample will work almost as well as using the entire data set, if the sample is **representative**
 - A sample is **representative** if it has approximately the same properties (of interest) as the original set of data

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Example: Precipitation in Australia

- This example is based on precipitation in Australia from the period 1982 to 1993.
 - The next slide shows
 - A histogram for the standard deviation of average monthly precipitation for 3,030 0.5° by 0.5° grid cells in Australia, and
 - A histogram for the standard deviation of the average yearly precipitation for the same locations.
- The average yearly precipitation has less variability than the average monthly precipitation.
- All precipitation measurements (and their standard deviations) are in centimeters.

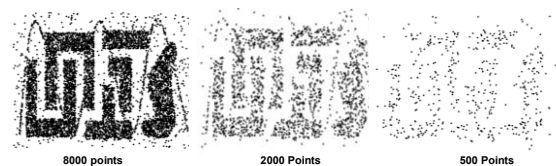
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Sampling

- Sampling is the main technique employed for data reduction.
 - It is often used for both the preliminary investigation of the data and the final data analysis.
- Statisticians often sample because **obtaining** the entire set of data of interest is too expensive or time consuming.
- Sampling is typically used in data mining because **processing** the entire set of data of interest is too expensive or time consuming.

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Sample Size



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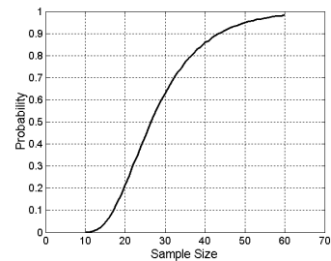
Types of Sampling

- Simple Random Sampling
 - There is an equal probability of selecting any particular item
 - Sampling without replacement
 - As each item is selected, it is removed from the population
 - Sampling with replacement
 - Objects are not removed from the population as they are selected for the sample.
 - In sampling with replacement, the same object can be picked up more than once
- Stratified sampling
 - Split the data into several partitions; then draw random samples from each partition

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Sample Size

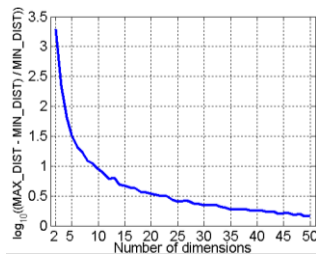
- What sample size is necessary to get at least one object from each of 10 equal-sized groups.



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Curse of Dimensionality

- When dimensionality increases, data becomes increasingly sparse in the space that it occupies
- Definitions of density and distance between points, which are critical for clustering and outlier detection, become less meaningful



- Randomly generate 500 points
- Compute difference between max and min distance between any pair of points

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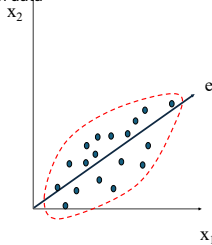
Dimensionality Reduction

- Purpose:
 - Avoid curse of dimensionality
 - Reduce amount of time and memory required by data mining algorithms
 - Allow data to be more easily visualized
 - May help to eliminate irrelevant features or reduce noise
- Techniques
 - Principal Components Analysis (PCA)
 - Singular Value Decomposition
 - Others: supervised and non-linear techniques

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Dimensionality Reduction: PCA

- Goal is to find a projection that captures the largest amount of variation in data



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Dimensionality Reduction: PCA

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Feature Subset Selection

- Another way to reduce dimensionality of data
- Redundant features
 - Duplicate much or all of the information contained in one or more other attributes
 - Example: purchase price of a product and the amount of sales tax paid
- Irrelevant features
 - Contain no information that is useful for the data mining task at hand
 - Example: students' ID is often irrelevant to the task of predicting students' GPA
- Many techniques developed, especially for classification

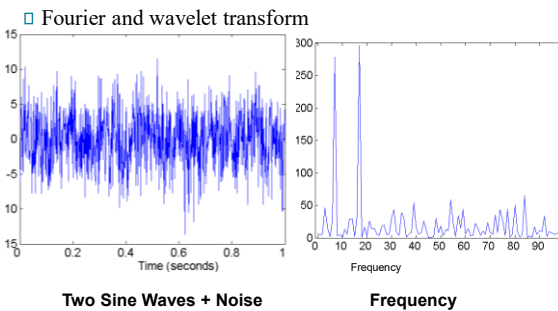
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Feature Creation

- Create new attributes that can capture the important information in a data set much more efficiently than the original attributes
- Three general methodologies:
 - Feature extraction
 - Example: extracting edges from images
 - Feature construction
 - Example: dividing mass by volume to get density
 - Mapping data to new space
 - Example: Fourier and wavelet analysis

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Mapping Data to a New Space



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Discretization

- **Discretization** is the process of converting a continuous attribute into an ordinal attribute
 - A potentially infinite number of values are mapped into a small number of categories
 - Discretization is commonly used in classification
- Many classification algorithms work best if both the independent and dependent variables have only a few values
- We give an illustration of the usefulness of discretization using the Iris data set

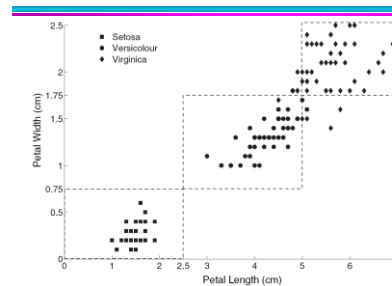
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Iris Sample Data Set

- Iris Plant data set.
 - Can be obtained from the UCI Machine Learning Repository (<http://www.uci.edu/learn/machine-learning-repository.html>)
 - From the statistician Douglas Fisher
 - Three flower types (classes):
 - Setosa
 - Versicolour
 - Virginica
 - Four (non-class) attributes
 - Sepal width and length
 - Petal width and length



Virginica. Robert H. Mohlenbrock. USDA NRCS. 1995. Northeast wetland flora: Field office guide to plant species. Northeast National Technical Center, Chester, PA. Courtesy of USDA NRCS Wetland



Petal width low or petal length low implies Setosa.
Petal width medium or petal length medium implies Versicolour.
Petal width high or petal length high implies Virginica.

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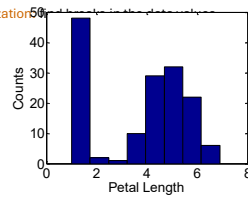
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Discretization: Iris Example ...

- How can we tell what the best discretization is?

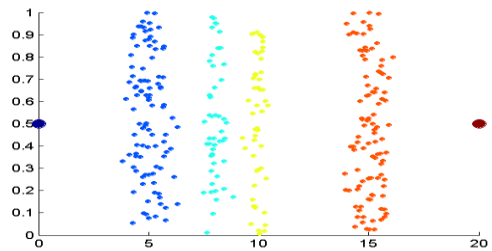
- Unsupervised discretization

- Example:
Petal Length



- Supervised discretization: Use class labels to find breaks

Discretization Without Using Class Labels

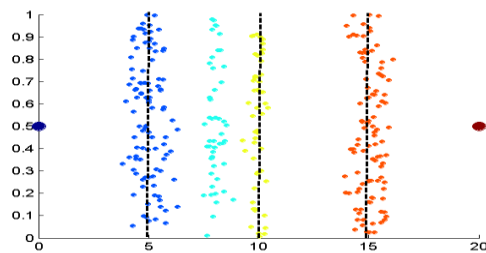


Data consists of four groups of points and two outliers. Data is one-dimensional, but a random y component is added to reduce overlap.

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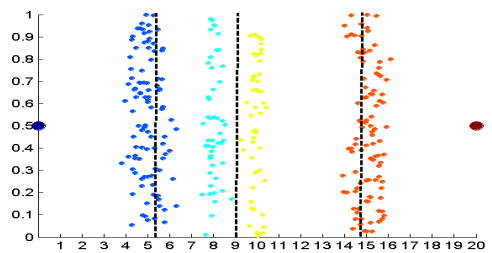
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Discretization Without Using Class Labels



Equal interval width approach used to obtain 4 values.

Discretization Without Using Class Labels

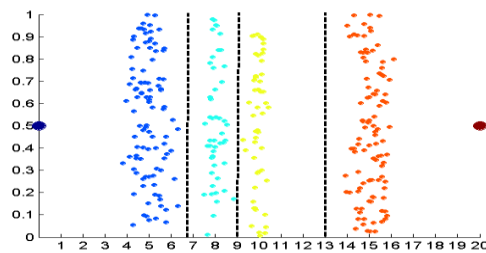


Equal frequency approach used to obtain 4 values.

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Discretization Without Using Class Labels



K-means approach to obtain 4 values.

Binarization

- Binarization maps a continuous or categorical attribute into one or more binary variables
- Typically used for association analysis
- Often convert a continuous attribute to a categorical attribute and then convert a categorical attribute to a set of binary attributes
 - Association analysis needs asymmetric binary attributes
 - Examples: eye color and height measured as {low, medium, high}

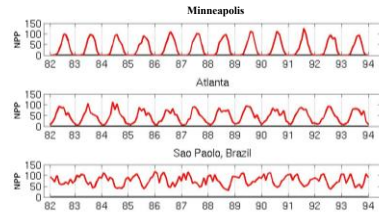
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Attribute Transformation

- An **attribute transform** is a function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values
 - Simple functions: x^k , $\log(x)$, e^x , $|x|$
 - Normalization**
 - Refers to various techniques to adjust to differences among attributes in terms of frequency of occurrence, mean, variance, range
 - Take out unwanted, common signal, e.g., seasonality
 - In statistics, **standardization** refers to subtracting off the means and dividing by the standard deviation

Example: Sample Time Series of Plant Growth



Net Primary Production (NPP) is a measure of plant growth used by ecosystem scientists.

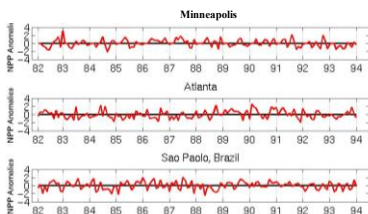
Correlations between time series

	Minneapolis	Atlanta	Sao Paolo
Minneapolis	1.0000	0.7591	-0.7581
Atlanta	0.7591	1.0000	-0.5739
Sao Paolo	-0.7581	-0.5739	1.0000

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Seasonality Accounts for Much Correlation



Normalized using monthly Z Score:
Subtract off monthly mean and divide by monthly standard deviation

Correlations between time series

	Minneapolis	Atlanta	Sao Paolo
Minneapolis	1.0000	0.0492	0.0906
Atlanta	0.0492	1.0000	-0.0154
Sao Paolo	0.0906	-0.0154	1.0000

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